

**VISVESVARAYA TECHNOLOGICAL
UNIVERSITY “JNANA SANGAMA”**



BELAGAVI-590018, KARNATAKA

A Project Report

On

LASER SECURITY IN CROP PROTECTION FROM WILD ANIMALS

By

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LASER SECURITY IN CROP PROTECTION FROM WILD ANIMALS

PROBLEM STATEMENT

Agriculture is the primary source of livelihood for a large portion of rural communities. However, farmers often face the persistent challenge of protecting their crops from wild animals. Animals such as deer, monkeys, wild boars, and even elephants frequently enter farmlands in search of food, causing extensive damage to crops. This not only reduces the agricultural yield but also leads to significant economic losses for farmers, many of whom operate on narrow profit margins. In some regions, repeated crop losses can push families into financial instability, affecting their ability to afford basic necessities, education for their children, and healthcare.

Traditional methods of crop protection, such as scarecrows, noise-makers, basic fencing, or manual surveillance by watchmen, have proven largely ineffective. Many of these methods require continuous human effort, are costly to maintain, and may not provide sufficient deterrence against larger or more intelligent animals. Moreover, the inefficiency of these solutions can lead to frustration and stress for farmers, negatively impacting their mental well-being and overall quality of life.

Another critical aspect of the problem is the conflict between humans and wildlife. When wild animals invade farmland, confrontations can occur, sometimes resulting in harm to the animals or to humans. Such conflicts not only raise ethical and safety concerns but can also lead to legal issues for farmers, especially when the animals involved are protected species. Communities living near forests or wildlife corridors are particularly vulnerable, as these encounters tend to be frequent and unpredictable.

The social issue, therefore, is not just economic but also ethical and safety-related. Farmers need a solution that allows them to protect their crops without causing harm to animals or exposing themselves to danger. Addressing this problem

would help improve the livelihood and security of farming communities, reduce human-wildlife conflict, and promote sustainable agricultural practices.

OBJECTIVES

To design a low-cost laser-based security system using simple electronics suitable for rural agricultural applications.

- To minimize crop damage caused by wildlife intrusion through early detection and sound-based deterrence.
- To develop a user-friendly and low-maintenance solution that requires minimal technical expertise to install and operate.
- To evaluate the system's reliability and effectiveness under real farm conditions, including day/night operation and weather variations.
- To promote the use of smart, sustainable technologies in agriculture for improving farm productivity and reducing economic losses.

Crop guarding is a common practice for rural farmers to protect their crops from wild animals like elephants and pigs. It also deters human predators and thieves. Farmers engage in crop guarding to stay vigilant and combat boredom, often using noise-making techniques like drums and crackers. Electric fences are used to keep elephants away, but if damaged, animals can freely enter farmland. To prevent this, poles can be camouflaged to avoid detection by animals.

The paper and study introduce a water system to increase water supply for efficient irrigation. Sensor estimates are sent to an Arduino Uno microcontroller for computation. Crop protection is ensured through voice and motion detection, emitting high-frequency sounds to deter insects, pests, and small animals. Additionally, a low-frequency sound emitter protects

crops from pests and bugs that emerge at night.

METHODOLOGY

Aim of the study

The aim of this study is to develop a system for preventing and detecting animal intrusion on farms using **laser**-ultrasound-based and Internet of Things (IoT) technology.

Research Design

This study will follow an experimental research design to assess the efficacy of the proposed **laser**-based and IoT system in preventing and detecting animal intrusion.

Instruments Used

Laser, Light Dependent Resistor (LDR), NodeMCU, Ultrasonic Alarm.

The Model

Laser -

Output Power - 100Mw and Output Wavelength - 532nm

Ultrasonic Alarm -

The following circuit was used for creating an alarm to generate ultrasonic sound using – 555 Timer IC (NE555), Resistors (20 kilo ohm and rheostat at 2 kilo ohm), Capacitor (2nF), Transducer (ultrasonic speaker).

Sensing Module -

The sensing module consists of an LDR connected to a NodeMCU housed in a 3-D printed casing.

Casing -

The following shows the design of the casing to house the electrical components.

EXPECTED OUTCOMES

The primary tangible output of this project will be a **laser-based crop protection system** that creates an invisible boundary around farmland. When an animal attempts to cross this boundary, the system will detect the intrusion and trigger an immediate alert, either through an alarm, lights, or notification to the farmer. This system is expected to reduce crop damage significantly, helping farmers protect their yield without the need for constant manual supervision. The installation of such a system can be scaled to fit small or large farms, making it adaptable to diverse agricultural settings.

Beyond the technical output, the project is expected to generate several important social impacts. Firstly, by reducing crop losses, farmers' incomes and financial stability will improve, directly enhancing their quality of life and economic resilience. Secondly, because the system deters animals without harming them, it promotes **human-wildlife coexistence**, reducing conflicts and ethical concerns associated with traditional deterrent methods. Farmers can protect their crops without resorting to harmful or aggressive measures, contributing to environmental and wildlife conservation efforts.

Additionally, the project could increase awareness among rural communities about the use of modern technology for sustainable agriculture. Adoption of such systems can reduce stress and mental burden on farmers, who often face anxiety over potential crop destruction. In the long term, successful implementation could inspire broader innovation in agricultural safety and security technologies, benefiting the community at large.

Overall, the project is expected to provide a **practical, cost-effective, and socially responsible solution** that safeguards crops, supports farmer livelihoods, and encourages harmony between humans and wildlife.

ALIGNMENT WITH SDGs

The proposed laser-based crop protection system aligns with several United Nations Sustainable Development Goals (SDGs) by addressing key social, economic, and environmental challenges faced by farming communities:

1. **SDG 2 – Zero Hunger:** By reducing crop losses caused by wild animals, the project helps increase agricultural productivity and ensures food security. Protecting crops supports farmers in producing more reliable yields, which contributes to the availability and accessibility of food.
2. **SDG 1 – No Poverty:** Crop damage often leads to financial instability for small and marginal farmers. By preventing losses, this system contributes to stable incomes, reducing poverty levels and improving livelihoods in rural communities.
3. **SDG 8 – Decent Work and Economic Growth:** Ensuring better crop protection allows farmers to focus on productive activities rather than constantly guarding their fields. This promotes economic growth in agricultural communities and supports sustainable rural development.

4. **SDG 12 – Responsible Consumption and Production:** The project encourages sustainable farming practices by minimizing waste due to damaged crops. Farmers can produce more efficiently without harming wildlife, contributing to responsible production methods.
5. **SDG 15 – Life on Land:** The system provides a non-lethal method to protect crops, reducing conflicts between humans and wildlife. This helps preserve biodiversity and promotes the protection of terrestrial ecosystems.
6. **SDG 9 – Industry, Innovation, and Infrastructure:** By integrating modern technology into traditional farming practices, the project fosters innovation in agricultural security solutions and strengthens infrastructure for sustainable agriculture.

Overall, this project not only addresses immediate agricultural challenges but also contributes to broader sustainable development objectives by improving farmer livelihoods, promoting coexistence with wildlife, and encouraging technological innovation in rural areas.

TENATIVE TIMELINE

ACTIVITY	EXPECTED COMPLETION DATE
Approval of the synopsis by the committee.	Tuesday, 30/09/2025
Review	Saturday, 31/10/2025
Project demonstration with the presentation.	Tuesday, 23/12/2025

REFERENCES

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Chicago- More, Amol. "Laser-Driven Smart Fence: A Novel Approach to Wildlife Deterrence in Agriculture."

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